

Parts of speech Tagging in NLP

In simple words, we can say that Pos tagging is a task of labelling each word in a sentence with its appropriate part of speech.

↓
Category of words having similar grammatical properties

Applications of Pos Tagging:

1. Named Entity Recognition
2. Question Answering System
3. Word Sense (disambiguation) disambiguation
4. Chatbots

Bayes Theorem

Dev

Weather	Mood
Sunny	Happy 😊
Rainy	Grumpy 😞

Straight forward,
clear distinction

↓
Lalitha

Adding some complexity Emission Prob's



Happy
0.8

Grumpy
0.2

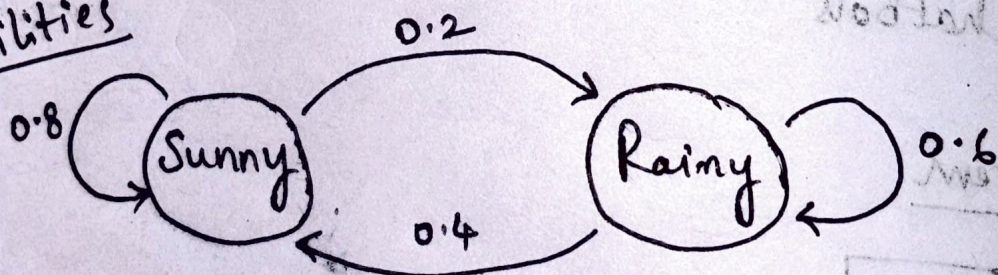
Happy
0.4

Grumpy
0.6

Der says the following:

Day 1	2	3	4	5	6	7
Happy	Grumpy	H	H	G	H	G
Sunny	Rainy	S	S	Rainy	S	R

Transition Probabilities

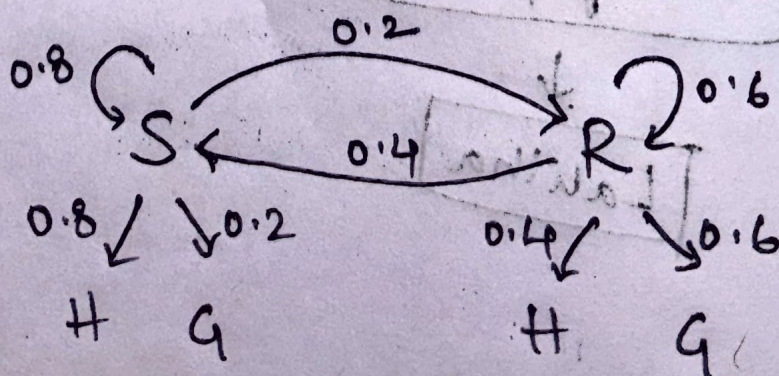


Observation: Can observe the mood of Der

Weather: Hidden state (Inferred)

based on observer (lalitha)

Hidden Markov Model



Questions

1. How did we find the probabilities?
2. What's the probability that a random day is S (or) R?
3. If Dev is H today, what's the prob that is S (or) R?
4. If for three days Dev is H, G, H, what was the weather?

Answers:

① observation

S S S S R R R S S S S R R S S S

Sunny

Rainy

S → S	8	0.8
S → R	2	0.2
R → S	2	0.4
R → R	3	0.6

Transition Probs

S S S S R R R S S S S R R S S S
G G G H H H H H H H H H H H

S → H	8	0.8
S → G	2	0.2
R → H	2	0.4
R → G	3	0.6

Emission Probs

②

S	10	$\frac{2}{3}$
R	5	$\frac{1}{3}$

Sunny (or) Rainy

$$S = 0.8 S + 0.4 R$$

$$S + R = 1$$

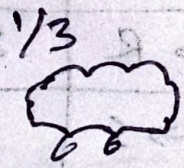
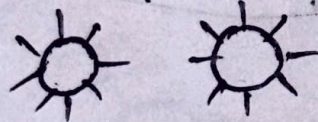
$$R = 0.2 S + 0.6 R$$

③

$P(H/S)$

$P(H/R)$

$\frac{2}{3}$



Friday

$$P(S) = \frac{2}{3}$$

$$P(R) = \frac{1}{3}$$

H	H
H	H
H	H
H	H

0.8

0.4

$$P(S/H) = \frac{8}{10}$$

$$P(R/H) = \frac{2}{10}$$

$$P(S/G) = \frac{2}{5}$$

$$P(R/G) = \frac{3}{5}$$

0.6

G	G
G	G
G	G
G	G

0.2

Thursday Friday

wednesday

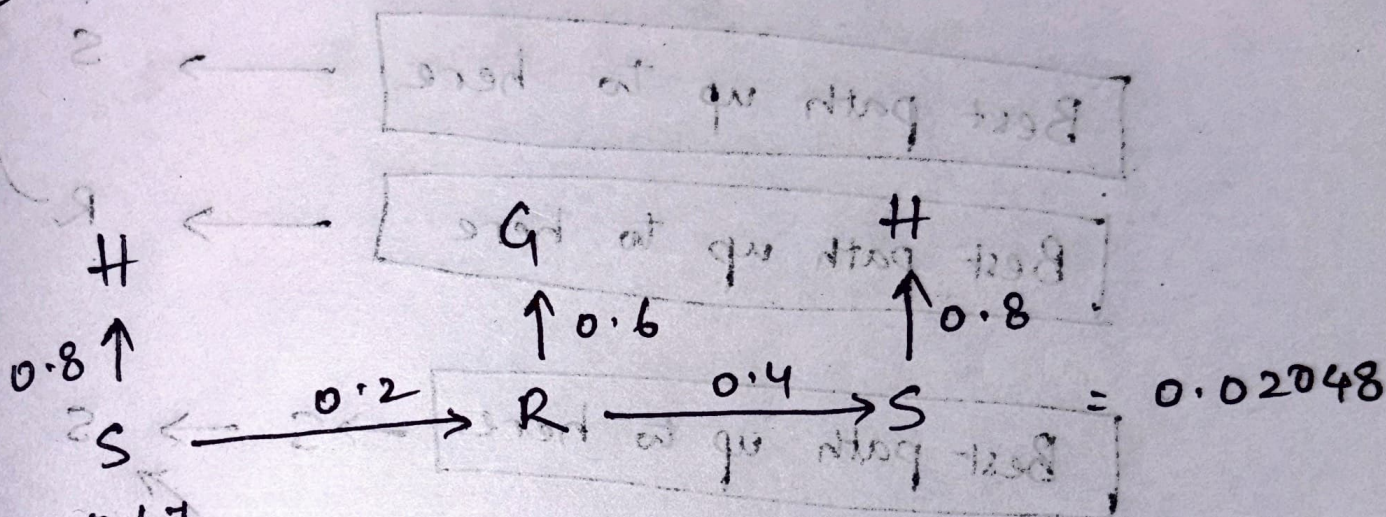
Thursday

Friday

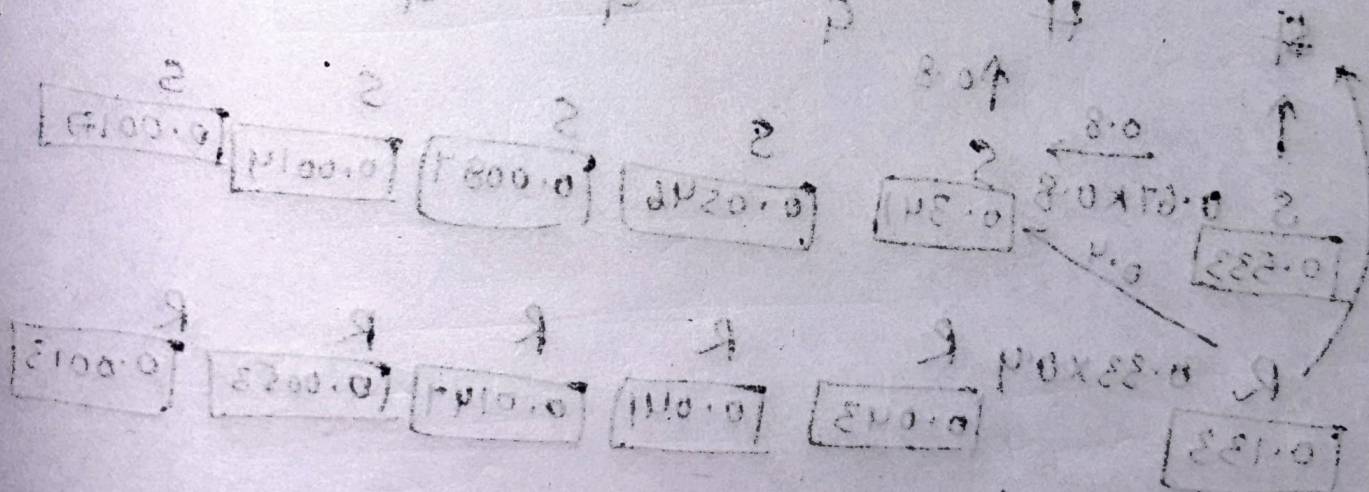
(H)
(S) (R)

Q
S R





n days $= 2^n$ combinations



Post-Genomic Gap & Transition Gap

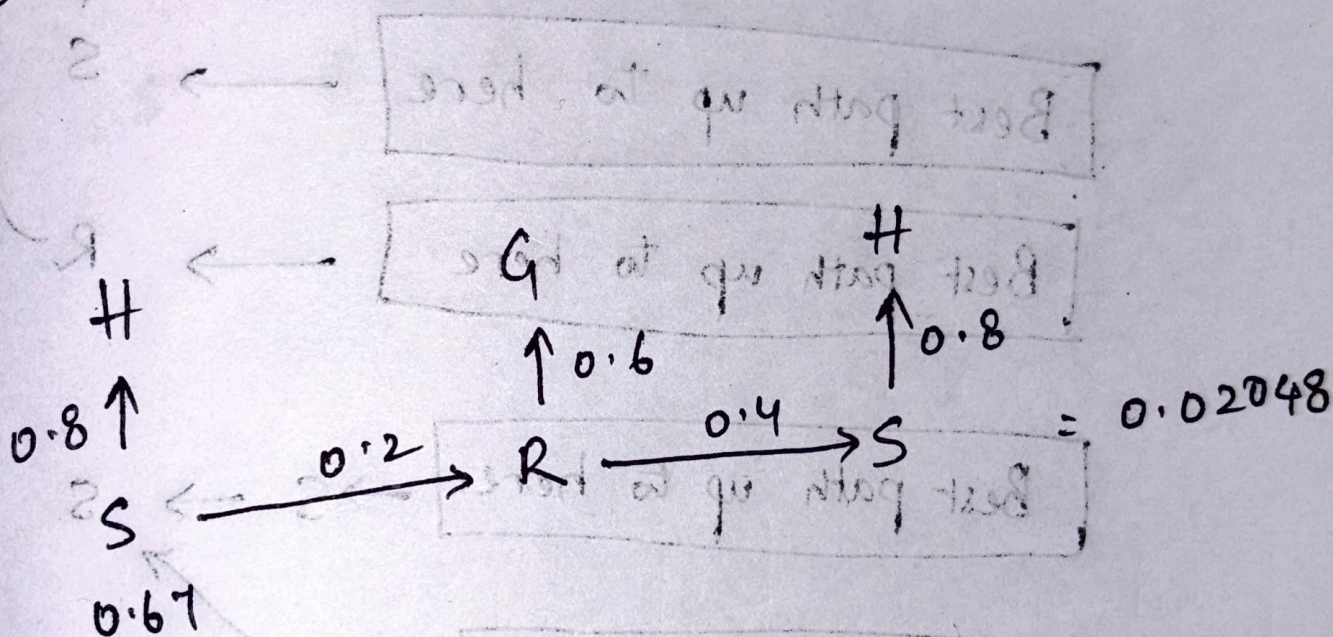
Weather History

Wednesday Thursday Friday

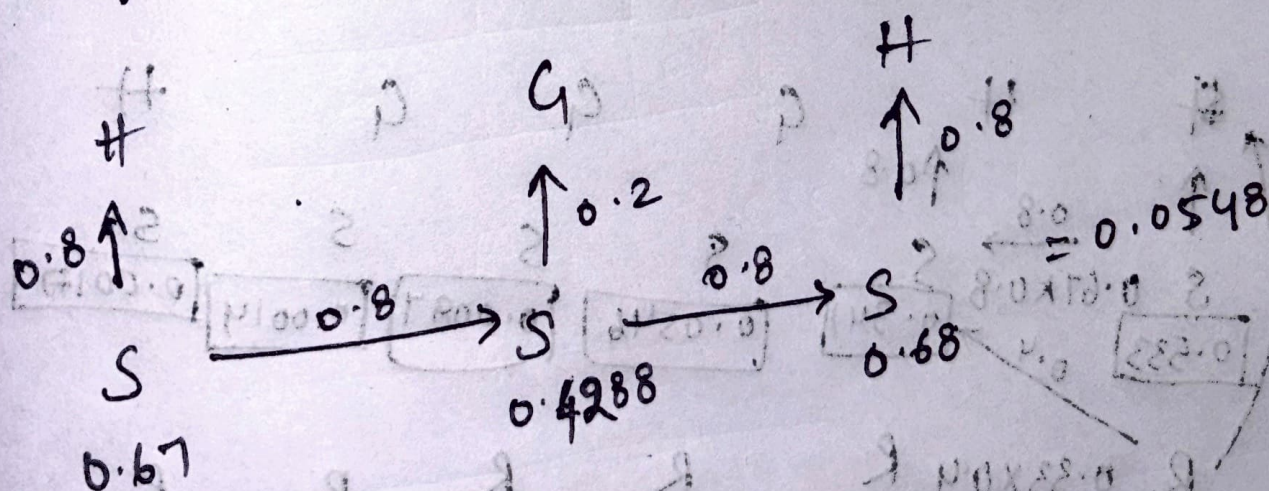
(H)
(S) (R)

(G)
(S) (R)

(H)
(S) (R)



n days $= 2^n$ combinations



But the above diagram is a simplified version of the actual weather history.

Viterbi Algorithm (Dynamic Programming)

Monday
H

T
H

W
G

Th
G

F
G

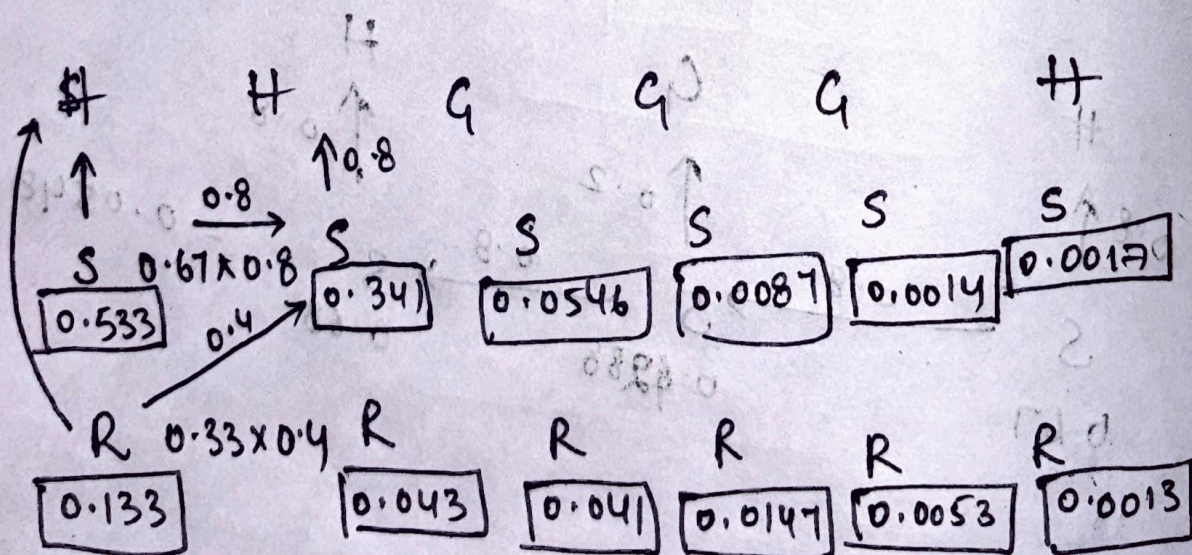
Saturday
H

Best path up to here → S

Best path up to here → R

Best path up to here → S → S

Best path up to here → R → R



best Previous prob $\times$ emission $\times$ transition prob

